Echo: team presentation guide

Group 6 | Companion to the existing ten-slide presentation | Preparation, not an extra assessed report

The board asks one question: **should Echo roll out aggressive push notifications to everyone?** Your answer is to keep normal notifications, because more app opens did not produce more listening or stronger later retention in the experiment.

Who presents what

Presenter	Slides and responsibility	Time
1	1-3: company, data, metrics and the claimed +30%	2:10
2 (you)	4-6: the A/B results, retention, channels and unit economics	2:35
3	7-10: model, backtest, scenarios, next test and recommendation	3:15
Total	Ten slides, one connected argument	8:00

This follows the slide timings already planned: **40, 45, 45, 55, 55, 45, 60, 55, 45 and 35 seconds.** Rehearse the transitions within those times. The jury question comes afterwards; its duration is not specified in the instructions received.

How to use this guide

Read page 2 together. It explains the vocabulary needed to follow the project. Then study your slide pages and the shared questions on pages 13-15.

Speak only the shaded scripts. The explanations underneath are for understanding and questions. Do not try to read the whole guide during the eight minutes. Adapt the wording to your own voice.

Explain the evidence. For each chart, say what is compared, what changes and why that matters for the decision. The notebook contains the detailed calculations; the presentation selects the main evidence.

Know the whole argument. The three presenters cover one opening section, one evidence section and one closing section. You are Presenter 2: explain the experiment, retention and channel economics. Everyone should understand the recommendation and the model assumptions.

Where to find things

Pages 3-12 follow slides 1-10 in order. Page 13 covers the live model question. Page 14 gives shared answers to likely questions. Page 15 explains how the work addresses the assignment and how to rehearse.

The vocabulary, without the course

Read together before dividing the speaking roles

Term	Meaning in this project
A/B experiment	Users are randomly assigned to two notification policies. Comparing the groups helps isolate the policy effect, subject to the assignment and measurement assumptions.
Session	An app open. Opening the app is not the same as listening for a long time.
Active user	Someone with at least one session in the week being measured.
Activated user	Someone with at least 20 cumulative listening minutes during the signup week. This does not require later retention.
W0 / W4 / W12	W0 is the Monday-Sunday signup week. W4 starts four weeks later; W12 starts twelve weeks later. These are calendar weeks, not exact individual seven-day windows.
Cohort	People grouped by signup time. The model uses weekly cohorts; some diagnostic tables group signups by month.
Retention	The share of the original eligible group with activity at a stated horizon. Everyone in that group stays in the denominator.
CAC	Customer acquisition cost: channel spend divided by new signups from that channel.
CPM	Revenue per 1,000 ad impressions. Echo earns approximately €9 per 1,000 impressions in the supplied case.
LTV / payback	LTV concerns value over a user lifetime. Payback is when accumulated value covers acquisition cost. Our observed revenues are gross and limited in time.
Backtest / forecast	A backtest predicts a held-out past period to check a model. A forecast calculates a future period under stated assumptions.
Scenario	A conditional what-if case. It is not a guaranteed outcome or a probability interval.

One example to keep the weeks clear

A signup on Wednesday 8 January 2025 belongs to W0, 6-12 January. W4 is 3-9 February. A session in W4 counts as a W4 return even if earlier weeks had no activity. The extract does not prove a permanent uninstall when a week is absent.

Three different ideas: activation asks whether someone first used the content; activity asks whether they used the app in a particular week; retention asks whether an original group returns later. Do not use the words interchangeably.

Slide 1 | Echo: notifications and growth

Presenter 1 | Target: 40 seconds | Read the script aloud; study the explanations separately

What to say

Echo is a free podcast app funded by advertising. The board asks whether aggressive push notifications should be rolled out to everyone. Our recommendation is to keep normal notifications. We found that aggressive notifications increase app opens, but reduce listening and later retention over a common twelve-week follow-up. We will explain the data, the experiment and a simple growth model checked against 2025. We will then show three future scenarios and propose a new, less intrusive experiment. Our decision focuses on real listening rather than opens alone.

What you are explaining

How Echo earns money. Users do not pay a subscription in this case. Listening generates ad impressions, and impressions generate estimated revenue. More sessions can help revenue, but the relationship is not automatic.

What the board wants. The aggressive version sends about five notifications per day with “Don't miss out!” copy. The board wants evidence on whether to roll this out. The control uses normal notifications.

What counts as evidence. The randomized experiment compares notification policies. The growth model and its scenarios examine economic assumptions. Those are different tools, and the scenarios do not replace the experiment.

What to point at. Use the recommendation on the title slide. Announcing the answer immediately helps the jury follow the evidence that comes next.

If the jury asks

Are you against all notifications? No. We recommend keeping normal notifications. We reject a rollout of the tested aggressive version and propose a separate test of a capped alternative.

Transition: Stay with Presenter 1: explain what was measured and how the data was prepared.

Slide 2 | Data and metrics

Presenter 1 | Target: 45 seconds | Read the script aloud; study the explanations separately

What to say

The data covers January 2025 to June 2026. After cleaning, we have 20,785 users and 332,736 activity rows. We removed exact duplicates and excluded 300 suspicious or impossible listening durations, while keeping the other information on those rows. Missing paid spend for May 2025 was estimated and flagged. We define activation as at least twenty minutes during the signup week. Our North Star is total weekly listening minutes, read alongside active users and retention. These definitions distinguish opening the app, consuming content and coming back later.

What you are explaining

An activity row is not a user. One user can appear in many different weeks. Removing duplicate rows prevents double counting; repeated activity across different weeks is legitimate.

Why twenty minutes? Ten minutes classified almost everyone as activated. Two sessions measure opens; adding twenty minutes to two sessions changed only 55 classifications in the comparison. Twenty minutes alone is simple and linked to content. It is a convention, not a proven predictor of retention.

What was kept. We removed 124 duplicate user rows and 1,663 duplicate activity rows. The 300 durations became missing values, not zero. Users with missing ages were retained as unknown.

What was estimated. May 2025 paid spend is €1,115, the average of April (€1,080) and June (€1,150). It is flagged. Costs for organic and podcast channels are unknown, not assumed zero.

If the jury asks

Why total listening as the North Star? It reflects consumption of the core audio product. We also track active users, median listening and retention, because totals can rise simply when the audience grows.

Transition: Stay with Presenter 1: examine what the Growth team may mean by the claimed +30%.

Slide 3 | Sessions during recruitment

Presenter 1 | Target: 45 seconds | Read the script aloud; study the explanations separately

What to say

The Growth team says sessions are up thirty percent. We reconstructed activity for experiment participants from 15 January to 15 March 2026. Control has 21,178 sessions and aggressive has 27,852: about 31.5 percent more in total. This makes the claim plausible, but we cannot prove that the team used exactly this method. These are calendar-period totals during recruitment, so participants have different follow-up lengths. To judge lasting value, we next compare every assigned user over the same twelve complete weeks after their signup week.

What you are explaining

How to read the bars. The vertical axis is total app opens, not listening minutes or distinct users. The percentage is (27,852 / 21,178 - 1) × 100, approximately 31.5%.

Why the dates need a caveat. Activity is weekly. The first included row covers 12-18 January, but relevant users signed up on or after 15 January. We include it assuming activity cannot occur before account creation. The final row covers 9-15 March.

Why the next slide uses another comparison. A January signup has more time to accumulate sessions during recruitment than a March signup. Equal user age and a per-assigned-user denominator give a more interpretable follow-up comparison.

What the result does not mean. The +31.5% is not proof of higher listening, stronger retention or profitability. It also is not a simple before-versus-after growth rate.

If the jury asks

Did we prove the team made a biased calculation? No. We found a plausible reconstruction of the claim. Its exact method is unknown. We explain why session totals alone are insufficient for the rollout decision.

Transition: To Presenter 2: “We have a plausible short-period increase in opens. Now let us compare lasting outcomes over equal follow-up.”

Slide 4 | Twelve weeks of follow-up

Presenter 2 | Target: 55 seconds | Read the script aloud; study the explanations separately

What to say

We compare all 1,366 control users with all 1,343 aggressive users, including people who became inactive. Each person has twelve complete follow-up weeks after the signup week. Sessions rise from 29.07 to 31.60 per person, an increase of 8.71 percent. However, known listening falls from 680.20 to 644.51 minutes, a decrease of 5.25 percent. The listening difference is negative even after allowing for sampling uncertainty, and missing-duration checks give the same reading. Estimated ad revenue is slightly higher, but its difference remains uncertain. More opens therefore do not demonstrate more lasting value.

What you are explaining

The correct denominator. First sum each participant's twelve-week outcomes, then average across everyone originally assigned to the group. Comparing only survivors would remove users the policy may have caused to stop returning.

The two charts use different units. The left chart counts sessions per person; the right counts minutes per person. Both groups use the same horizon. The relative changes divide each aggressive-minus-control difference by the control mean.

Why we say known listening. Twenty durations are missing within follow-up. Their true minutes are unknown. We show known totals, check comparable-value estimates, and temporarily exclude the twenty affected users as a separate sensitivity analysis. Both checks support lower listening.

What uncertainty means here. The mean listening difference is -35.69 minutes, with a 95% bootstrap interval from -62.83 to -9.61. This interval stays below zero. The ad revenue difference interval includes zero, so a positive point estimate is not a demonstrated gain.

If the jury asks

Was the experiment only two months long? Recruitment and assignment ran from 15 January to 15 March. Data through June allows all assigned users twelve complete follow-up weeks. Assignment dates and follow-up duration are different.

Transition: Stay with Presenter 2: show whether users keep returning during those twelve weeks.

Slide 5 | Retention declines

Presenter 2 | Target: 55 seconds | Read the script aloud; study the explanations separately

What to say

This chart follows the share of all assigned users who are active in each week. The horizontal axis is weeks after the signup week, not calendar months. The groups are close early on, and the week-four difference is uncertain. The gap grows later. At week twelve, 51.3 percent of control users return, compared with 39.6 percent in aggressive. That is about 11.71 percentage points less. Together with lower cumulative listening, this argues against a rollout. Notification fatigue is a possible explanation, but satisfaction and uninstall behavior were not measured directly.

What you are explaining

Read a point as a proportion. At W12, 51.3% means roughly 51 of every 100 originally assigned control users have activity in that week. Each user counts once for that week, no matter how many sessions they have.

Points versus percent. 51.3% minus 39.6% is about 11.7 percentage points. A relative decline would divide that gap by 51.3%, giving roughly 22.8%. The slide reports points, not the relative decline.

What a return requires. A user needs at least one session in that particular week. They do not need to have been active every earlier week. We are not counting everyone who ever returns after W4.

What the evidence supports. The W12 difference interval runs from -15.48 to -8.10 points. The later retention measures support the main listening result. They are secondary outcomes with intervals unadjusted for multiple comparisons.

If the jury asks

Does inactivity mean an uninstall? No. It means no recorded activity in that week. We cannot infer permanent churn, uninstall or dissatisfaction directly from this extract.

Transition: Stay with Presenter 2: “The experiment raises a retention concern. We also need to understand how channels and user revenue affect growth.”

Slide 6 | Channels and unit economics

Presenter 2 | Target: 45 seconds | Read the script aloud; study the explanations separately

What to say

Paid social has lower week-four retention than the other channels, even after comparing a similar mix of signup months and platforms. This remains an association, not proof that the channel causes the difference. For unit economics, we use the same January-to-June 2025 signups with complete revenue horizons. Paid users generate about €1.62 of ad revenue over fifty-two weeks, against €2.61 acquisition cost. Mean payback is therefore not reached at this horizon. Costs for organic and podcast channels are unknown, and these revenues are gross rather than profit.

What you are explaining

What adjusted retention does. Calculate channel retention within signup month × platform groups, then average with the same weights across channels. This asks what the rates would look like with a common observed mix. It does not control every possible difference.

The table's two populations. Adjusted W4 retention uses signups with complete W4 follow-up across the extract. The 52-week revenue comparison uses January-June 2025 signups. The columns should not be treated as measurements on one identical population.

The revenue calculation. Observed ad impressions × approximately €9 / 1,000. Average revenue includes inactive cohort members. Paid is €1.615, organic €2.554 and podcast €2.946 at 52 weeks.

Why this is not full LTV. LTV concerns a complete lifetime. We observe only fixed horizons and lack operating costs and margins. Not recovering CAC by 52 weeks does not tell us an exact future payback date or prove it will never be reached.

If the jury asks

Why might paid users be less loyal? An ad could attract a curiosity-driven trial, while a store search or podcast promotion could reflect existing audio interest. Motivations are unobserved, so this is a hypothesis rather than a finding.

Transition: To Presenter 3: “These channel differences and acquisition costs feed our model. We will now show how the model works and what it predicts.”

Slide 7 | Model and 2025 backtest

Presenter 3 | Target: 60 seconds | Read the script aloud; study the explanations separately

What to say

Our model forecasts weekly cohorts separately by channel. Next active users equal existing active users times a continuation rate, plus new signups. Paid signups come from budget divided by CAC. Organic and podcast follow their historical signup trends. An internal validation selected one continuation rate per channel: listening error was 1.57 percent, versus 2.54 percent for the tested age bands. We then froze the structure and predicted the second half of 2025 using parameters available before July and actual budgets as inputs. Listening error was 3.85 percent, versus 14.27 percent for a linear trend. The model increasingly underestimates activity; we retain that limitation rather than retuning on the test.

What you are explaining

A simple example, not an Echo estimate. With 100 active users, 95% weekly continuation and 20 new signups, next week has 100 × 0.95 + 20 = 115 predicted active users. Weekly cohorts keep track of signup timing; the chosen continuation rate is constant within each channel.

What the chart shows. Actual and forecast weekly listening during 30 June-28 December 2025. Here W1-W26 labels the 26 backtest weeks, not a single cohort's age. Units are thousands of listening minutes per week.

What 3.85% means. Add absolute weekly forecast errors and divide by total actual listening. This is WAPE. It is not an error bound for every week, and it is not 96.15% certainty about the future.

What was known. Parameters stop at 29 June 2025. Actual held-out budgets are supplied, so this tests outcomes given those budgets, not budget prediction. The last week forecasts 4,844 active users against 5,229 observed.

If the jury asks

Why not a more complex model? The specific age-band version tested had more internal validation error. We keep the simpler option. This does not mean every complex model would be worse.

Transition: Stay with Presenter 3: “With the model checked against the past, we can now show what different future assumptions imply.”

Slide 8 | Three six-month scenarios

Presenter 3 | Target: 55 seconds | Read the script aloud; study the explanations separately

What to say

These projections cover twenty-six weeks, from 29 June to 27 December 2026. All start from 7,468 observed active users, keep normal notifications and use a €1,630 monthly paid budget. The base case retains historical parameters. The optimistic case reduces CAC and churn probability by five percent; the pessimistic case increases both by five percent. Listening totals are 16.49, 16.97 and 16.05 million minutes respectively. These are conditional outcomes, not guaranteed gains or confidence intervals. They show how acquisition efficiency and retention affect growth; the notification decision itself comes from the randomized experiment.

What you are explaining

Read the chart and table differently. The lines show listening per week, in thousands of minutes. The table sums all 26 weeks, in millions of minutes. In this chart W1-W26 labels forecast weeks, not cohort age.

What changes and what stays fixed. Both CAC and churn probability change together. Budget, organic/podcast signup trends, mean consumption and CPM remain fixed. These cases do not isolate the contribution of each changed parameter.

Five percent is relative. For illustration, a 10% departure probability reduced by 5% becomes 9.5%, not 5%. Since continuation = 1 - departure probability, we change departures rather than subtracting five points from continuation.

The model's starting point. The stock is observed in the week of 22-28 June 2026. Behavior parameters remain those validated on 2025. Forecasts therefore assume historical behavior applies under normal notifications; they do not measure future outcomes.

If the jury asks

Why not baseline, targeted and full notification rollout? The course uses base/optimistic/pessimistic economic scenarios. Policy rollout scenarios are another useful approach, but targeting has no demonstrated beneficial segment here, and applying new-user effects to all existing users requires assumptions.

Transition: Stay with Presenter 3: explain what a safer, genuinely new notification experiment would test.

Slide 9 | Next experiment plan

Presenter 3 | Target: 45 seconds | Read the script aloud; study the explanations separately

What to say

We propose a new experiment rather than assuming a gentler policy will work. Control keeps normal notifications. The variant caps notifications at three per week, uses neutral copy and offers user frequency settings. Eligible new signups are randomized after enabling notifications. We measure listening per assigned user over eighty-four days, with a retention guardrail limiting deterioration to two percentage points. Initial sample planning suggests about 7,800 users per group, driven by the guardrail. This estimate needs a pilot check. Twelve weeks is each person's follow-up; total recruitment and follow-up will take longer.

What you are explaining

The variant is a proposal. Three per week, neutral copy and user choice have not been tested in the existing data. We cannot transfer the aggressive policy's measured effect to this new version.

Primary outcome and guardrail. The primary outcome is cumulative listening per assigned user. A guardrail is a condition that must also pass, to avoid accepting a listening gain with an unacceptable retention loss. The proposed margin is two percentage points.

Why so many participants? An approximate calculation uses 80% power and a 5% significance level. A 5% listening effect requires about 1,877 users per group, while the retention guardrail requires about 7,723. We round up to 7,800 and would revisit this after an instrumentation pilot.

The decision rule. Look for a positive listening effect with an interval excluding zero and a passed retention guardrail. Monitor opt-outs and complaints. Do not repeatedly stop at a favorable interim result. Daily instrumentation is needed for individual 84-day follow-up.

If the jury asks

Can the test finish in twelve weeks? Not from recruitment start. Each participant needs 84 days of follow-up, including the last person enrolled. Eligible signup volume and costs determine how long recruitment takes.

Transition: Stay with Presenter 3: close by answering the board's question in one sentence.

Slide 10 | Recommendation for Echo

Presenter 3 | Target: 35 seconds | Read the script aloud; study the explanations separately

What to say

Our recommendation is to keep normal notifications and not roll out the tested aggressive version. The increase in app opens is plausible, but equal follow-up shows less listening and lower later retention, without a demonstrated ad revenue gain. A separate capped-frequency experiment is the next step. The model helps us understand economic assumptions, but does not turn them into observed facts. Our conclusion applies to the new users studied and the available follow-up. We choose listening and user return as the basis for the decision, rather than sessions alone.

What you are explaining

What to emphasize. Answer the actual board question. The recommendation is supported by listening, retention and revenue evidence together, not by one attractive percentage.

What you are not claiming. The study does not establish lifetime effects or results for existing users. We do not claim to know the psychological mechanism behind the decline.

What the model contributes. The forecast compares consequences of assumptions. Costs are incomplete, so it does not establish net profitability. The scenarios support economic reasoning rather than prove the notification recommendation.

How to finish. Stop after the recommendation and invite the question. The notebook should already be open and executed, with the scenario controls easy to find. Do not begin browsing code during the closing sentence.

If the jury asks

What would make you reconsider? A well-designed new experiment showing a credible listening gain, a passed retention guardrail and acceptable user feedback. The current data does not provide that evidence for aggressive notifications.

Transition: End of the eight-minute presentation. Move to the jury question; do not restart the slides.

Handling the live model question

After the eight-minute presentation | One operator, shared understanding

The jury may ask how an outcome changes when an assumption changes. Presenter 3 can operate the notebook and explain the scenarios. Presenter 2 can connect the answer to the experiment and unit economics. All three should understand the assumptions. This is a suggested team arrangement, not a requirement from the instructor.

Before the defense

1. Open the right notebook. Use G06_Echo_model.ipynb from the project root, with the data folder available and requirements installed. Run all cells once and confirm that results and charts appear. Code inputs are collapsed in compatible readers; expand the scenario controls.

2. Find the three scenario cells. They are the parameter cell, the results cell and the chart cell under “Three scenarios over twenty-six weeks”. Keep the submitted values in a separate unchanged copy.

Control	Current value	Meaning
BUDGET_PAID_MONTHLY	1630	Paid acquisition budget per month, in euros
SCENARIO_CHANGE	0.05	Relative change to both CAC and churn in the optimistic/pessimistic cases
FORECAST_WEEKS	26	How many future weeks the model calculates

Worked question: what if paid budget rises by 20%?

State the assumption. “We will hold CAC, retention and consumption fixed and increase only paid budget.” Set BUDGET_PAID_MONTHLY to 1956, then rerun the parameter, results and chart cells in order.

Read the recalculated outputs. Compare base-case last-week active users, cumulative listening, ad revenue and paid spend with the submitted base. Read the actual new results; do not invent a percentage for total active users.

Explain the mechanism. At unchanged CAC, new paid signups rise by 20%. Total active users need not rise by 20%, because existing cohorts and the other channels also contribute. Organic and podcast do not receive an automatic paid-ad uplift.

Give the limit. “This is conditional on CAC staying constant. The data does not prove that a higher real budget would preserve acquisition efficiency.” Restore 1630 afterwards.

When the question needs more than a control change

A churn-only change is not the same as changing SCENARIO_CHANGE, because that control changes both churn and CAC. A full notification rollout also is not an existing switch. Explain the extra assumptions before adapting the model. Do not claim a displayed economic scenario answers a different policy question.

Shared answers for the whole group

Use these for questions, not as extra material during the eight minutes

Experiment and measurement

Why compare all assigned users? Keeping original assignment avoids selecting only users who remained active. Weeks with no activity contribute zero usage; a missing duration on a recorded active week remains unknown. These are different situations.

What is a bootstrap confidence interval? Repeatedly sample users with replacement within each group and calculate the difference. Keep each user's weeks together. The interval describes sampling uncertainty under this method; it does not capture all missing-data or design uncertainty.

Does an interval crossing zero prove no effect? No. It means the estimate does not clearly establish the direction at the reported confidence level. That is why we say “no demonstrated revenue gain”, not “exactly no revenue effect”.

What do power and significance mean? Power is the chance that a planned test detects the target effect if it is real, under its assumptions. The proposed 80% is not an 80% probability that our conclusion is correct. A 5% significance level controls a specified false-positive rate; it is not the size of the effect.

Are cohort and calendar time the same? No. Cohort age measures weeks since signup. Calendar time is the actual date. Experiment retention uses cohort age. Backtest and scenario chart weeks label consecutive calendar forecast weeks.

Economics and model

Why not call €1.62 the LTV? It is observed gross ad revenue over 52 weeks for the stated 2025 signups. Lifetime duration and operating margins are unavailable. Calling it full lifetime value would overstate what was measured.

Did we use the course Excel calculator? We did not fill in its workbook. Our notebook applies the same growth principles using weekly cohorts, channel parameters and ad impressions. The course permits a spreadsheet or notebook. The supplied workbook remains unchanged as a reference.

Why is backtesting not predicting the future? A past period is held out as if its outcomes were unknown. Parameters are fitted earlier, then forecasts are compared with actual held-out outcomes. It checks whether the model is useful before applying future assumptions.

Why not recalibrate on the failed final weeks? That would use the final test to improve its own score. We keep the underestimation visible. A later model update would require a separate validation plan.

Are the ±5% assumptions measured improvements? No. They are explicit scenario choices. They change CAC and departures jointly. They do not claim that a particular product intervention achieves those changes.

Decision

Can we target a successful subgroup instead? Potentially, but a credible subgroup benefit has not been demonstrated here. Searching many small groups can produce chance results. A targeting recommendation would need its own evidence and validation.

What the jury should take away

Assignment coverage, rehearsal and sources

Course question	Where our work answers it
Funnel and loop	The notebook maps signup, activation and W4 return, with retention drop-off. The follow-notification-listen-return loop is plausible, not a measured referral engine.
Comparable segments	Channel, platform and age comparisons use common follow-up and composition adjustments. Channel findings are associations. Slide 6 selects the channel result.
Unit economics	CAC, retention, fixed-horizon ad revenue and gross payback are examined by cohort/channel where available. Full lifetime LTV and net profit remain unestablished.
Changes and causes	The notebook shows eighteen-month acquisition, activity and monetization patterns. The randomized notification comparison supplies the strongest causal evidence; other changes remain descriptive.
North Star	Total known weekly listening minutes, supported by active users, median listening and retention. Slides 2, 4 and 5 explain why opens alone are insufficient.

One rehearsal together

First pass: understanding. Each person explains their slides without reading the script. Teammates ask what the denominator, time window and units are. Check every use of “percent” and “percentage points”.

Second pass: eight minutes. Use the scripts as a starting point and time the whole team, including handovers. If you run long, simplify your spoken explanation rather than silently removing an analytical result. Do not read the study notes aloud.

Third pass: the model. Practice the +20% budget question, explain the assumptions, read the recalculated outputs and restore the original values. Everyone should be able to state the main model limitation.

Source trail for checking a number

G06_Echo_slides.pptx: the existing ten-slide presentation and speaker notes.
G06_Echo_report.pdf: the four-page report with main results and limits.
G06_Echo_model.ipynb: definitions, cleaning, cohort comparisons, experiment estimates, backtest and scenario controls.
docs/cleaning_log.csv: cleaning rules and affected counts.
Echo - Project data pack.pdf: fictional company, columns, random assignment dates and approximate CPM.
Growth modeling class.pdf: funnels, loops, North Star, unit economics and scenario planning.
IMT - Growth Model Simulator.xlsx: supplied teaching simulator; not the calculation engine used in our notebook.

This guide explains the current deliverables. Its toy examples illustrate concepts and are explicitly separated from measured Echo results. The detailed project brief and a confirmed Growth Modeling grading grid were not available; no extra rubric requirements are invented here.